04 - Backlog (ADO-Ready Epics, Features, Tasks)

Document Type: Operational
Phase: Planning
Audience: [project-managers, engineers, devops]
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Backlog Structure

This backlog follows Azure DevOps hierarchy: **Epic** → **Feature** → **Task**.
Each task includes clear acceptance criteria suitable for definition of done (DoD).

Estimation: T-shirt sizing (XS=1pt, S=2pt, M=3pt, L=5pt, XL=8pt)
Priority: P0 (Critical), P1 (High), P2 (Medium), P3 (Low)

Epic 1: FinOps Hubs Foundation

Goal: Establish storage infrastructure, event routing, and validation framework.
Business Value: Enable centralized cost data collection with automated ingestion triggers.
Effort: 13 points (Sprints 1-2)

Feature 1.1: ADLS Gen2 Landing Zone

User Story: As a FinOps analyst, I need structured storage containers to organize raw, processed, and archived cost data for lifecycle management.

Task 1.1.1: Create Storage Containers (XS)

Description: Create 4 containers in `marcosandboxfinopshub`: raw, processed, archive, checkpoint.

Acceptance Criteria:

- ☐ All 4 containers visible via `az storage container list`
- ☐ Container access level: Private (no anonymous access)
- ☐ Test file upload succeeds in each container
- ☐ Documentation: Screenshot of Portal Containers blade
- ☐ Evidence artifact: `tools/finops/out/storage_containers.json`

Effort: 1 point
Priority: P0
Assigned To: DevOps Engineer

Task 1.1.2: Configure Lifecycle Management Policy (S)

Description: Implement auto-tiering to Cool (90 days) and Archive (180 days) for cost optimization.

Acceptance Criteria:

- ☐ Bicep module `storage-lifecycle.bicep` deployed successfully
- ☐ Policy visible in Portal → Storage Account → Lifecycle management
- ☐ Rule count: 2 (MoveRawToCool, MoveArchiveToArchive)
- ☐ Test blob shows scheduled tier change in metadata
- ☐ Evidence: `az storage account management-policy show` output saved

Effort: 2 points
Priority: P1
Assigned To: Cloud Architect

Task 1.1.3: Migrate Existing Exports to Raw Container (M)

Description: Reorganize current `costs/` blobs into hierarchical `raw/costs/{subscription}/{YYYY}/{MM}/` structure.

Acceptance Criteria:

- ☐ Script `migrate-costs-to-raw.sh` creates copies in raw/ preserving hierarchy
- ☐ Validation: Row count spot-check for 5 sample CSVs (source vs. destination)
- ☐ Original `costs/` container renamed to `costs-old` (not deleted)
- ☐ Documentation: Migration log with blob count and total size

Effort: 3 points
Priority: P0
Assigned To: Data Engineer

Feature 1.2: Export Configuration Update

User Story: As a cost data engineer, I need exports to land in the correct raw/ container with proper event notifications.

Task 1.2.1: Update Cost Management Export Destinations (S)

Description: Reconfigure portal exports to target `raw/costs/{SubscriptionName}` path.

Acceptance Criteria:

- ☐ Both exports (EsDAICoESub-Daily, EsPAICoESub-Daily) updated in Portal
- ☐ Manual trigger succeeds and blob lands in `raw/costs/.../YYYY/MM/` path
- ☐ Export history shows updated root folder path
- ☐ Evidence: Screenshots of export configuration + successful run

Effort: 2 points
Priority: P0
Assigned To: FinOps Administrator

Feature 1.3: Event Grid Integration

User Story: As an automation engineer, I need blob creation events to trigger ingestion pipelines automatically.

Task 1.3.1: Verify Event Grid System Topic (S)

Description: Confirm existing system topic `marcosandboxfinopshub-52dd...` is active and receiving events.

Acceptance Criteria:

- ☐ System topic status: Succeeded
- ☐ Metrics show events in last 24 hours (after export runs)
- ☐ Event types: `Microsoft.Storage.BlobCreated` for `/raw/costs/` prefix
- ☐ Evidence: Event Grid Metrics screenshot showing event count

Effort: 2 points
Priority: P0
Assigned To: Cloud Architect

Task 1.3.2: Create Event Subscription to ADF (M)

Description: Wire Event Grid to trigger ADF pipeline `ingest-costs-to-adx` on blob creation.

Acceptance Criteria:

- ☐ Event subscription created: `finops-ingest-trigger`
- ☐ Filter: Prefix `/raw/costs/`, Suffix `.csv.gz`
- ☐ Test: Upload sample CSV → ADF pipeline triggered within 2 minutes
- ☐ Delivery success rate: 100% (check Event Grid Metrics)
- ☐ Evidence: Event subscription JSON + ADF run history

Effort: 3 points
Priority: P0
Assigned To: Data Engineer

Epic 2: Analytics & Ingestion

Goal: Deploy ADX cluster, create schema, and automate CSV ingestion from Blob to ADX.
Business Value: Enable large-scale cost analytics with KQL queries and Power BI integration.
Effort: 21 points (Sprints 3-4)

Feature 2.1: ADX Cluster Deployment

User Story: As a FinOps analyst, I need a scalable analytics platform to query millions of cost records with sub-second response time.

Task 2.1.1: Deploy ADX Cluster (Dev SKU) (M)

Description: Deploy `marcofinopsadx` cluster in canadacentral with Dev SKU (2 nodes, D11_v2).

Acceptance Criteria:

- ☐ Bicep module `adx-cluster.bicep` deploys without errors
- ☐ Cluster status: Running (Portal → Azure Data Explorer Clusters)
- ☐ Database `finopsdb` created with 31-day hot cache, 5-year retention
- ☐ Query editor accessible: `https://dataexplorer.azure.com/clusters/marcofinopsadx`
- ☐ Test query succeeds: `.show databases | where DatabaseName == "finopsdb"`
- ☐ Evidence: Cluster URI, database name, principal ID output saved

Effort: 3 points
Priority: P0
Assigned To: Cloud Architect

Task 2.1.2: Create ADX Schema and Tables (L)

Description: Execute KQL script to create `raw_costs`, `apim_usage` tables, materialized view, and ingestion mappings.

Acceptance Criteria:

- ☐ Tables created: `.show tables` returns 2 rows (raw_costs, apim_usage)
- ☐ Ingestion mapping: `.show table raw_costs ingestion csv mappings` returns CostExportMapping
- ☐ Materialized view: `.show materialized-views` returns normalized_costs
- ☐ Function: `.show functions` returns AllocateCostByApp
- ☐ Policies applied: partitioning (by Date), retention (5 years), update (IngestionTime)
- ☐ Evidence: Screenshots of KQL command results

Effort: 5 points
Priority: P0
Assigned To: Data Architect

Task 2.1.3: Test ADX Ingestion with Sample CSV (S)

Description: Validate ingestion mapping by manually uploading 5-row sample CSV.

Acceptance Criteria:

- ☐ Sample CSV ingested using `.ingest inline` or ADF test run
- ☐ Query validation: `raw_costs | count` returns 5
- ☐ Column types validated: `raw_costs | getschema` matches expected types
- ☐ No parsing errors in `.show ingestion failures`
- ☐ Evidence: ADX query results screenshot

Effort: 2 points
Priority: P0
Assigned To: Data Engineer

Feature 2.2: ADF Pipeline Development

User Story: As a data engineer, I need automated pipelines to decompress, validate, and ingest cost CSVs into ADX without manual intervention.

Task 2.2.1: Create Managed Identity for ADF (XS)

Description: Deploy user-assigned managed identity `mi-finops-adf` with required RBAC roles.

Acceptance Criteria:

- ☐ Managed identity created: `az identity show -n mi-finops-adf`
- ☐ Role 1: Storage Blob Data Contributor on marcosandboxfinopshub
- ☐ Role 2: ADX Database Ingestor on finopsdb
- ☐ Role assignments validated: `az role assignment list --assignee <principalId>`
- ☐ ADX principal check: `.show database finopsdb principals` includes mi-finops-adf
- ☐ Evidence: Role assignment outputs saved

Effort: 1 point
Priority: P0
Assigned To: Cloud Architect

Task 2.2.2: Create ADF Linked Services (S)

Description: Configure connections to Blob Storage and ADX using managed identity authentication.

Acceptance Criteria:

- ☐ Linked service: `ls_marcosandbox_blob` (Azure Blob Storage, managed identity auth)
- ☐ Linked service: `ls_marcofinops_adx` (Azure Data Explorer, managed identity auth)
- ☐ Test connection succeeds for both linked services
- ☐ Evidence: Screenshots of linked service configurations

Effort: 2 points
Priority: P0
Assigned To: Data Engineer

Task 2.2.3: Create ADF Datasets (S)

Description: Define datasets for source (Blob CSV) and sink (ADX table).

Acceptance Criteria:

- ☐ Dataset: `ds_blob_cost_csv` (DelimitedText, gzip compression, parameterized blobUrl)
- ☐ Dataset: `ds_adx_raw_costs` (AzureDataExplorer, table=raw_costs, mapping=CostExportMapping)
- ☐ Preview data validates schema for both datasets
- ☐ Evidence: Dataset JSON definitions exported

Effort: 2 points
Priority: P0
Assigned To: Data Engineer

Task 2.2.4: Implement Pipeline: ingest-costs-to-adx (L)

Description: Build pipeline with decompression, validation, and copy activities.

Acceptance Criteria:

- ☐ Pipeline activities: Get Metadata → Copy Data → Update Checkpoint
- ☐ Parameters: blobUrl (string, passed from Event Grid)
- ☐ Copy activity: Source=ds_blob_cost_csv, Sink=ds_adx_raw_costs, Enable staging
- ☐ Error handling: Retry policy (3 attempts), failure alert (email/webhook)
- ☐ Test run with sample blob succeeds
- ☐ ADX validation: Row count matches CSV (minus header)
- ☐ Evidence: Pipeline JSON + successful debug run screenshot

Effort: 5 points
Priority: P0
Assigned To: Data Engineer

Task 2.2.5: Implement Pipeline: backfill-historical (L)

Description: Create pipeline to process 500+ historical blobs (12 months) in parallel.

Acceptance Criteria:

- ☐ Pipeline activities: Get Metadata (list blobs) → ForEach (parallelism=5) → Execute ingest pipeline
- ☐ Progress tracking: Update checkpoint every 100 blobs
- ☐ Manual trigger succeeds and completes within 4 hours
- ☐ Validation: Daily row counts match expected (spot-check 10 dates)
- ☐ Evidence: Pipeline run history + ADX validation query results

Effort: 5 points
Priority: P1
Assigned To: Data Engineer

Epic 3: APIM Attribution & Telemetry

Goal: Capture cost attribution headers from APIM and correlate with Azure spend.
Business Value: Enable chargeback/showback to consuming applications and cost centers.
Effort: 13 points (Sprints 5-6)

Feature 3.1: APIM Policy Configuration

User Story: As a FinOps analyst, I need to know which applications are driving Azure spend so I can allocate costs accurately.

Task 3.1.1: Implement APIM Cost Attribution Policy (M)

Description: Add base policy to inject/normalize `x-eva-costcenter`, `x-eva-caller-app`, `x-eva-environment` headers.

Acceptance Criteria:

- ☐ Policy added to "All APIs" base policy in Portal
- ☐ Inbound section extracts headers from JWT claims or request headers
- ☐ Default values applied for missing headers (UNKNOWN)
- ☐ Test request with curl includes header: `x-costcenter: TEST123`
- ☐ Policy XML exported and saved to repo: `infra/apim/base-policy.xml`
- ☐ Evidence: Screenshot of policy editor + test request with headers

Effort: 3 points
Priority: P0
Assigned To: API Developer

Task 3.1.2: Configure APIM Diagnostics to App Insights (S)

Description: Enable Application Insights logging for all APIs with 100% sampling.

Acceptance Criteria:

- ☐ Logger created: marco-sandbox-appinsights linked to APIM
- ☐ Diagnostics enabled: All APIs, verbosity=information, sampling=100%
- ☐ Custom dimensions captured: x-eva-costcenter, x-eva-caller-app, x-eva-environment
- ☐ Test request generates log entry in App Insights within 60 seconds
- ☐ KQL query validates: `requests | where customDimensions["x-eva-costcenter"] != ""`
- ☐ Evidence: App Insights query screenshot showing custom dimensions

Effort: 2 points
Priority: P0
Assigned To: DevOps Engineer

Feature 3.2: Telemetry Ingestion to ADX

User Story: As a data engineer, I need APIM request logs in ADX to join with cost data by timestamp and resource.

Task 3.2.1: Create ADF Pipeline: ingest-apim-telemetry (L)

Description: Build pipeline to export App Insights logs and ingest into ADX `apim_usage` table.

Acceptance Criteria:

- ☐ Pipeline scheduled: Daily 3 AM UTC
- ☐ Data source: App Insights Analytics query (yesterday's requests)
- ☐ Sink: ADX table `apim_usage` with columns: Timestamp, ApiId, CallerApp, CostCenter, etc.

- ☐ First run succeeds and populates ADX: `apim_usage | count` > 0
- ☐ Evidence: Pipeline JSON + ADX query result showing sample rows

Effort: 5 points
Priority: P0
Assigned To: Data Engineer

Task 3.2.2: Validate Allocation Function (M)

Description: Test `AllocateCostByApp()` KQL function with sample data.

Acceptance Criteria:

- ☐ Sample data: 100 APIM requests, 10 cost records for APIM resource
- ☐ Function execution succeeds: `AllocateCostByApp() | take 10`
- ☐ Validation: Total AllocatedCost == Total APIM cost (within 1% tolerance)
- ☐ Results show per-app breakdown with RequestCount and AllocatedCost columns
- ☐ Evidence: ADX query result screenshot

Effort: 3 points
Priority: P1
Assigned To: Data Analyst

Epic 4: Reporting & Visualization

Goal: Deploy Power BI reports and dashboards for executive cost visibility.
Business Value: Enable self-service cost analytics and trend monitoring.
Effort: 8 points (Sprint 7)

Feature 4.1: Power BI Workspace Setup

User Story: As a FinOps manager, I need interactive dashboards to monitor spending trends and identify optimization opportunities.

Task 4.1.1: Create Power BI Workspace (XS)

Description: Provision shared workspace `FinOps Analytics` with Pro licensing.

Acceptance Criteria:

- ☐ Workspace created: `FinOps Analytics` (shared capacity or Premium)
- ☐ Members added: FinOps team (Contributor), Executives (Viewer)
- ☐ Evidence: Workspace members list screenshot

Effort: 1 point
Priority: P1
Assigned To: BI Administrator

Task 4.1.2: Deploy Cost Trend Dashboard (M)

Description: Create PBIX with ADX DirectQuery connection showing daily/monthly cost trends.

Acceptance Criteria:

- ☐ Data source: ADX cluster `marcofinopsadx`, database `finopsdb`, table `normalized_costs`
- ☐ Visuals: Line chart (90-day trend), cards (MTD spend, budget %), bar chart (top 10 RGs)
- ☐ Filters: Subscription, Date range, Resource Group
- ☐ Refresh: Auto (DirectQuery, no scheduled refresh needed)
- ☐ Published to workspace and pinned to dashboard
- ☐ Evidence: Dashboard screenshot + sample KQL queries used

Effort: 3 points
Priority: P1
Assigned To: BI Developer

Task 4.1.3: Deploy APIM Allocation Report (M)

Description: Create report showing cost allocation by caller app and cost center.

Acceptance Criteria:

- ☐ Data source: ADX function `AllocateCostByApp()`
- ☐ Visuals: Table (CallerApp, Cost, Requests), Pie chart (by CostCenter)
- ☐ Date range parameter: Last 30 days (default)
- ☐ Published to workspace
- ☐ Evidence: Report screenshot with sample allocation data

Effort: 3 points
Priority: P1
Assigned To: BI Developer

Task 4.1.4: Deploy Tag Compliance Report (S)

Description: Create report showing untagged resources and compliance percentage.

Acceptance Criteria:

- ☐ Data source: ADX table `normalized_costs`
- ☐ Visuals: Gauge (compliance %), table (untagged resources with cost)
- ☐ Filter: Date = Yesterday (latest full day)
- ☐ Published to workspace
- ☐ Evidence: Report screenshot showing compliance percentage

Effort: 2 points
Priority: P2
Assigned To: BI Developer

Epic 5: Governance & Hardening

Goal: Enforce policies, enable private endpoints, and implement CI/CD for infrastructure.
Business Value: Reduce security risk and improve compliance posture.
Effort: 13 points (Sprints 8-9)

Feature 5.1: Azure Policy Enforcement

User Story: As a security architect, I need to enforce tagging standards to prevent cost allocation gaps.

Task 5.1.1: Deploy Policy: Require CostCenter Tag (S)

Description: Create and assign custom policy that denies resource creation without CostCenter tag.

Acceptance Criteria:

- ☐ Policy definition JSON created: `require-costcenter-tag.json`
- ☐ Policy assigned to scope: `/subscriptions/.../resourceGroups/EsDAICoE-Sandbox`
- ☐ Test: Create resource without tag → Deployment fails with policy error
- ☐ Compliance scan shows non-compliant resources (if any)
- ☐ Evidence: Policy assignment + test deployment failure

Effort: 2 points
Priority: P2
Assigned To: Cloud Architect

Feature 5.2: Private Endpoint Implementation

User Story: As a security engineer, I need private connectivity to storage and ADX to eliminate public internet exposure.

Task 5.2.1: Deploy VNet for FinOps Resources (M)

Description: Create VNet `vnet-finops-canadacentral` with private endpoint subnet.

Acceptance Criteria:

- ☐ VNet created: Address space 10.100.0.0/16
- ☐ Subnet: `subnet-privatelink` (10.100.1.0/24, private endpoint policies disabled)
- ☐ VNet peering (if needed): Connect to existing DevBox VNet
- ☐ Evidence: VNet properties screenshot

Effort: 3 points
Priority: P2
Assigned To: Network Engineer

Task 5.2.2: Enable Private Endpoint for Storage (M)

Description: Configure private endpoint for `marcosandboxfinopshub` blob service.

Acceptance Criteria:

- ☐ Private endpoint created: `pe-marcosandbox-blob`
- ☐ Private DNS zone: privatelink.blob.core.windows.net
- ☐ Storage firewall: Deny public access, allow VNet subnet only
- ☐ Test from DevBox: `az storage blob list` succeeds
- ☐ Test from public IP: Connection refused
- ☐ Evidence: Private endpoint status + network test results

Effort: 3 points
Priority: P2
Assigned To: Network Engineer

Task 5.2.3: Enable Private Endpoint for ADX (M)

Description: Configure private endpoint for `marcofinopsadx` cluster.

Acceptance Criteria:

- ☐ Private endpoint created: `pe-marcofinops-adx`
- ☐ Private DNS zone: privatelink.canadacentral.kusto.windows.net
- ☐ ADX public access: Disabled
- ☐ Test from DevBox: KQL query succeeds
- ☐ Evidence: Private endpoint status screenshot

Effort: 3 points
Priority: P2
Assigned To: Network Engineer

Feature 5.3: CI/CD Automation

User Story: As a DevOps engineer, I need automated deployment pipelines to avoid manual errors and ensure consistency.

Task 5.3.1: Create GitHub Actions Workflow (S)

Description: Implement workflow to validate and deploy Bicep templates on PR/merge.

Acceptance Criteria:

- ☐ Workflow file: `.github/workflows/deploy-finops.yml`
- ☐ Triggers: PR to main (validate only), merge to main (validate + deploy)
- ☐ Jobs: Bicep build, ARM validation, deployment to EsDAICoE-Sandbox
- ☐ Secrets configured: AZURE_CREDENTIALS (service principal)
- ☐ Test run: PR triggers validation, merge triggers deployment
- ☐ Evidence: Workflow run history screenshot

Effort: 2 points
Priority: P2

Assigned To: DevOps Engineer

Sprint Planning Summary

Sprint	Epics/Features	Total Points	Key Deliverables
Sprint 1-2	Epic 1 (Foundation)	13	Storage hierarchy, Event Grid wired
Sprint 3-4	Epic 2 (Analytics)	21	ADX deployed, ingestion pipeline live
Sprint 5-6	Epic 3 (Attribution)	13	APIM headers, telemetry in ADX
Sprint 7	Epic 4 (Reporting)	8	Power BI dashboards published
Sprint 8-9	Epic 5 (Governance)	13	Policies enforced, private endpoints enabled
Total	**5 Epics**	**68 points**	**Full FinOps Hubs implementation**

Velocity Assumption: 15 points/sprint (2-week sprints)
Estimated Timeline: 9 weeks (4.5 sprints)

Success Metrics (KPIs)

Metric	Target	Measured By
Cost Data Completeness	100% (12 months backfilled)	ADX query: daily row counts
Ingestion Latency	<30 minutes (export → ADX)	ADF pipeline duration
APIM Attribution Coverage	>80% of requests have CostCenter	App Insights query: % with custom dimension
Tag Compliance	>90% of resources tagged	ADX query: untagged resource count
Cost Allocation Accuracy	±5% (APIM allocated vs. actual)	Validation query comparing allocation to invoice
Power BI Query Performance	❤️ seconds (for 90-day trend)	Power BI Performance Analyzer

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Change Log

Date	Author	Change
2026-02-17 08:20 AM ET	Marco Presta	Initial ADO-ready backlog with epics, features, and tasks

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Document Status: Ready for sprint planning
Next Review: Sprint 0 planning session (2026-02-20)